

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim:- Creating graphical reports using ,ggplot2 (R).

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

Source
Console Terminal Background Jobs
> library(ggplot2)
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> ai_jobs <- read_csv("ai_jobs.csv")
> str(ai_jobs)
'data.frame': 50000 obs. of 14 variables:
 $ job_id      : chr "01FD0TVBDIVU" "ZMF8MD04V30T" "CX1945NQ4FMY" "QJ7YHLIC320C" ...
 $ job_title   : chr "MLOps Engineer" "Data Analyst" "MLOps Engineer" "Applied Scientist"
 $ company_type : chr "Research Lab" "Startup" "Research Lab" "Research Lab" ...
 $ industry    : chr "Education" "Education" "Tech" "Healthcare" ...
 $ country     : chr "Australia" "Germany" "Canada" "Australia" ...
 $ city        : chr "Remote" "Remote" "Remote" "Remote" ...
 $ remote_type : chr "Remote" "Remote" "Remote" "Remote" ...
 $ experience_level : chr "Entry" "Entry" "Senior" "Entry" ...
 $ min_experience_years : int 0 0 5 0 2 5 5 5 0 0 ...
 $ salary_min_usd : int 56873 54803 149980 53483 102977 148201 143946 147177 61851 57350 ...
 $ salary_max_usd : int 72223 85599 175806 86477 127298 179967 173591 171959 83711 69593 ...
 $ employment_type : chr "Full-time" "Full-time" "Full-time" "Full-time" ...
 $ posted_year   : int 2023 2024 2021 2023 2023 2021 2020 2023 2024 2023 ...
 $ company_size  : chr "Large" "Medium" "Large" "Medium" ...
> head(ai_jobs)
  job_id      job_title company_type industry country city
1 01FD0TVBDIVU MLOps Engineer Research Lab Education Australia Remote
2 ZMF8MD04V30T Data Analyst Startup Healthcare Germany Remote
3 CX1945NQ4FMY MLOps Engineer Research Lab Tech Canada Remote
4 QJ7YHLIC320C Applied Scientist Research Lab Healthcare Australia Remote
5 F070PVM9ER14 Machine Learning Engineer Research Lab Finance Australia Sydney
6 9T84AZTJXEP Data Analyst Startup Healthcare USA Seattle

Air: Poor Tomorrow
```

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

Source
Console Terminal Background Jobs
> ggplot(ai_jobs, aes(x = experience_level)) +
+   geom_bar() +
+   labs(
+     title = "Job Distribution by Experience Level",
+     x = "Experience Level",
+     y = "Number of Jobs"
+   ) +
+   theme_minimal()
> ggplot(ai_jobs, aes(x = remote_type)) +
+   geom_bar() +
+   labs(
+     title = "Jobs by Work Mode",
+     x = "Work Mode",
+     y = "Number of Jobs"
+   ) +
+   theme_minimal()
> ggplot(ai_jobs, aes(x = experience_level, y = salary_max_usd)) +
+   geom_boxplot() +
+   labs(
+     title = "Salary Distribution by Experience Level",
+     x = "Experience Level",
+     y = "Maximum Salary (USD)"
+   ) +
+   theme_minimal()
> ggplot(ai_jobs, aes(x = min_experience_years, y = salary_max_usd)) +
```

NAME:- CHETAN MANDAVKAR

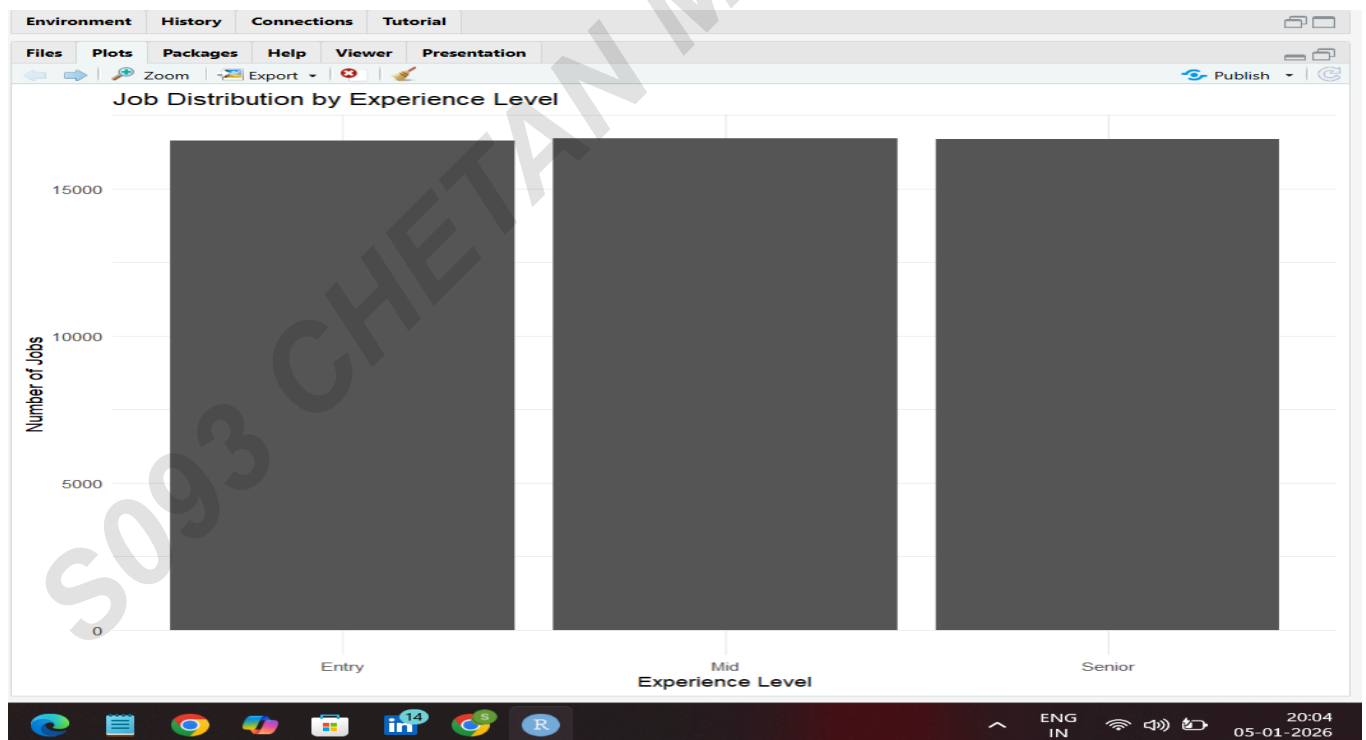
ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
R - R4.5.2 - ~/R
+ labs(title = "Jobs by Work Mode",
+ x = "Work Mode",
+ y = "Number of Jobs")
+ theme_minimal()
> ggplot(ai_jobs, aes(x = experience_level, y = salary_max_usd)) +
+ geom_boxplot() +
+ labs(
+ title = "Salary Distribution by Experience Level",
+ x = "Experience Level",
+ y = "Maximum Salary (USD)"
+ )
+ theme_minimal()
> ggplot(ai_jobs, aes(x = min_experience_years, y = salary_max_usd)) +
+ geom_point() +
+ labs(
+ title = "Experience vs Maximum Salary",
+ x = "Minimum Experience (Years)",
+ y = "Maximum Salary (USD)"
+ )
+ theme_minimal()
> ggplot(ai_jobs, aes(x = industry)) +
+ geom_bar() +
+ labs(
+ title = "Number of Jobs by Industry",
+ x = "Industry",
+ y = "Number of Jobs"
+ )
+ theme_minimal() +
+ theme(axis.text.x = element_text(angle = 45, hjust = 1))
> ggplot(ai_jobs, aes(x = posted_year)) +
+ geom_bar() +
+ labs(
+ title = "AI Jobs Posted Per Year",
+ x = "Year",
+ y = "Number of Jobs"
+ )
+ theme_minimal()
>

Environment History Connections Tutorial
R - Global Environment
Data
AI_Impact_on_Jobs_2 3000 obs. of 18 variables
ai_job_market 2000 obs. of 12 variables
ai_jobs 50000 obs. of 14 variables
anova_model Large aov (13 elements, 1.4 MB)
big_companies 2000 obs. of 12 variables
body_weight_df 60 obs. of 2 variables
brain_weight_df 59 obs. of 2 variables
car_delivery_sorted 47 obs. of 10 variables
chi_result List of 9
correlation_matrix num [1:18, 1:18] 1 0.014168 0.005821 -0.000325 -0.0109...
delivery 18 obs. of 3 variables
delivery_multi_sort 18 obs. of 4 variables
delivery_sorted_dis... 200 obs. of 10 variables
delivery_sorted_key... 18 obs. of 4 variables
delivery_sorted_name 18 obs. of 3 variables
delivery_sorted_time 200 obs. of 10 variables
df 7691 obs. of 10 variables
diabetes_data 10000 obs. of 20 variables
energy 5000 obs. of 6 variables
Food_Delivery_Route... 200 obs. of 10 variables
high_cost 1101 obs. of 6 variables
high_cost_low_occ 118 obs. of 6 variables
high_cost_not_se 495 obs. of 6 variables
high_salary 1093 obs. of 12 variables
high_salary_senior 336 obs. of 12 variables
industry_filter 0 obs. of 12 variables
jobs 2000 obs. of 12 variables
low_occupants_filter 1386 obs. of 6 variables
Files Plots Packages Help Viewer Presentation
Air Poor Tomorrow
Search
ENG IN 20:03 05-01-2026
```

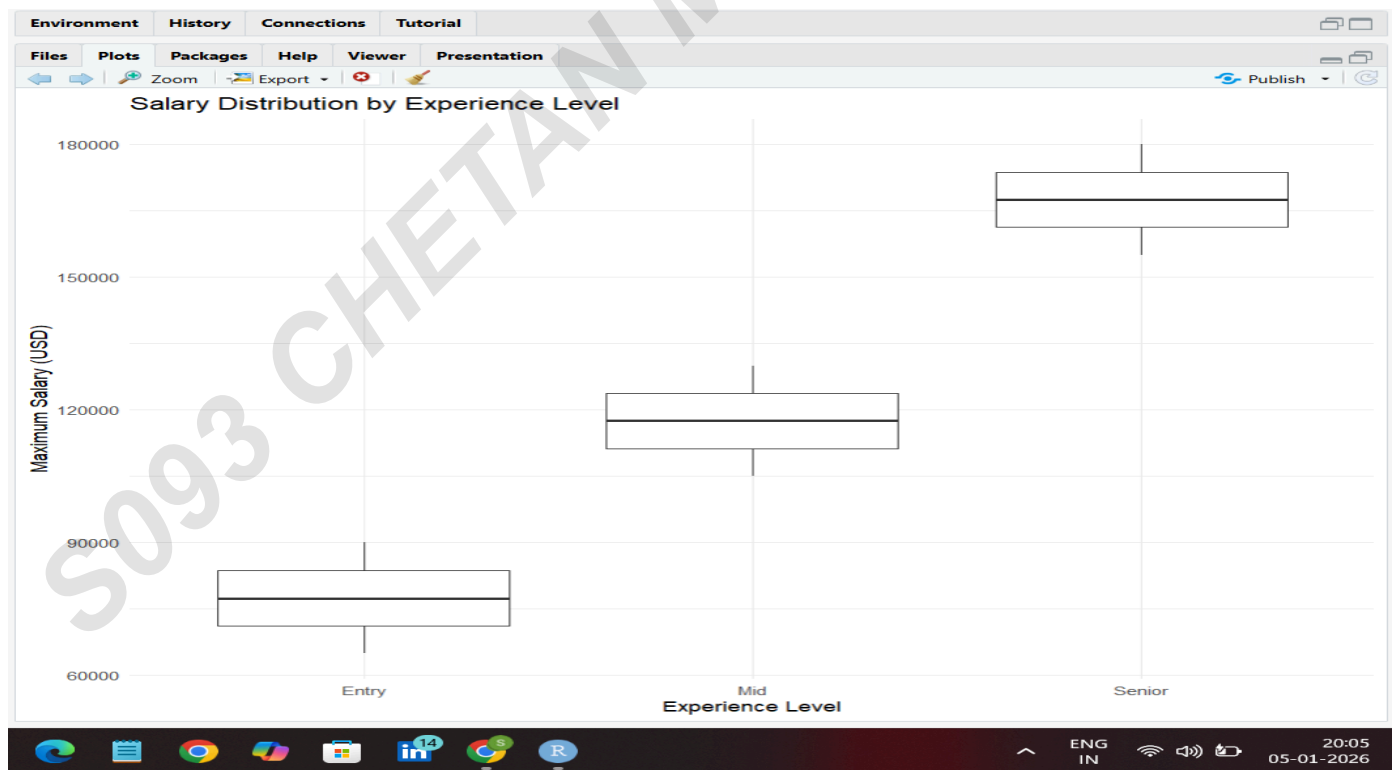
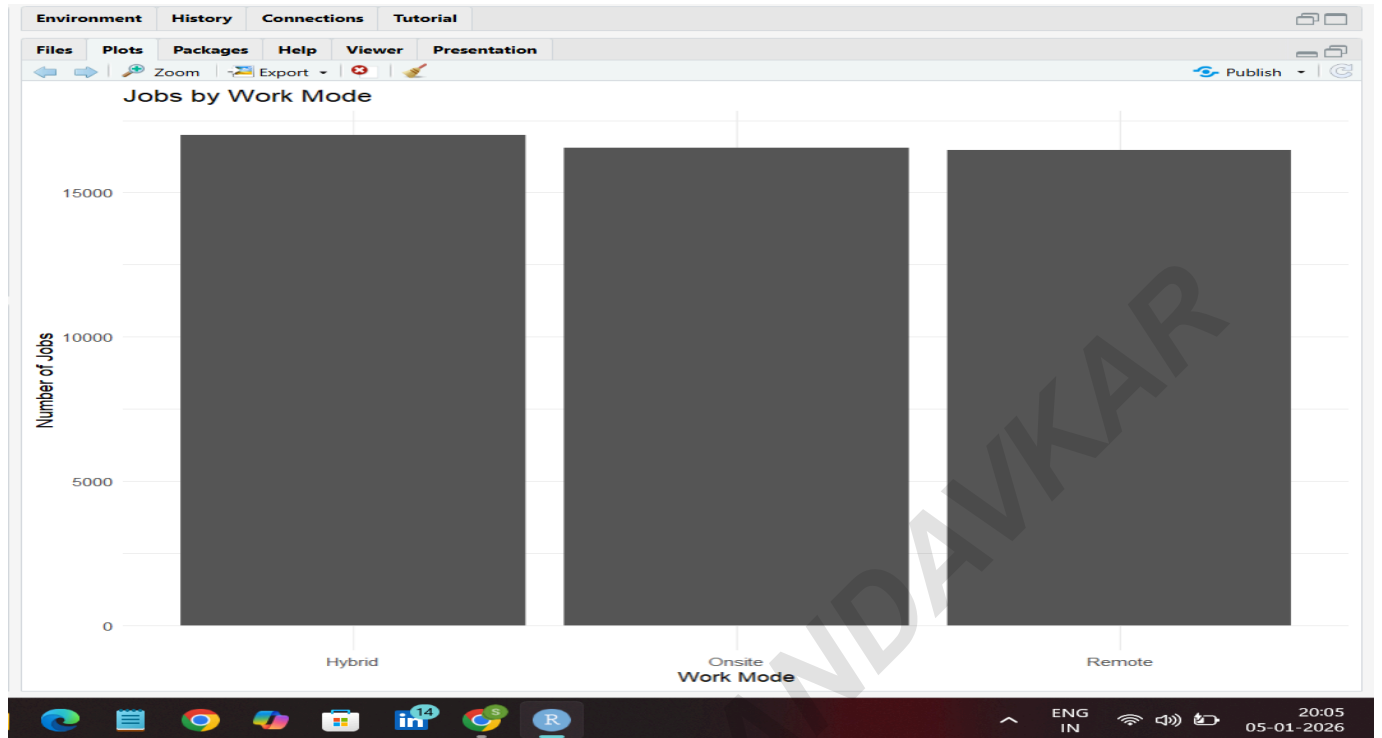


NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE

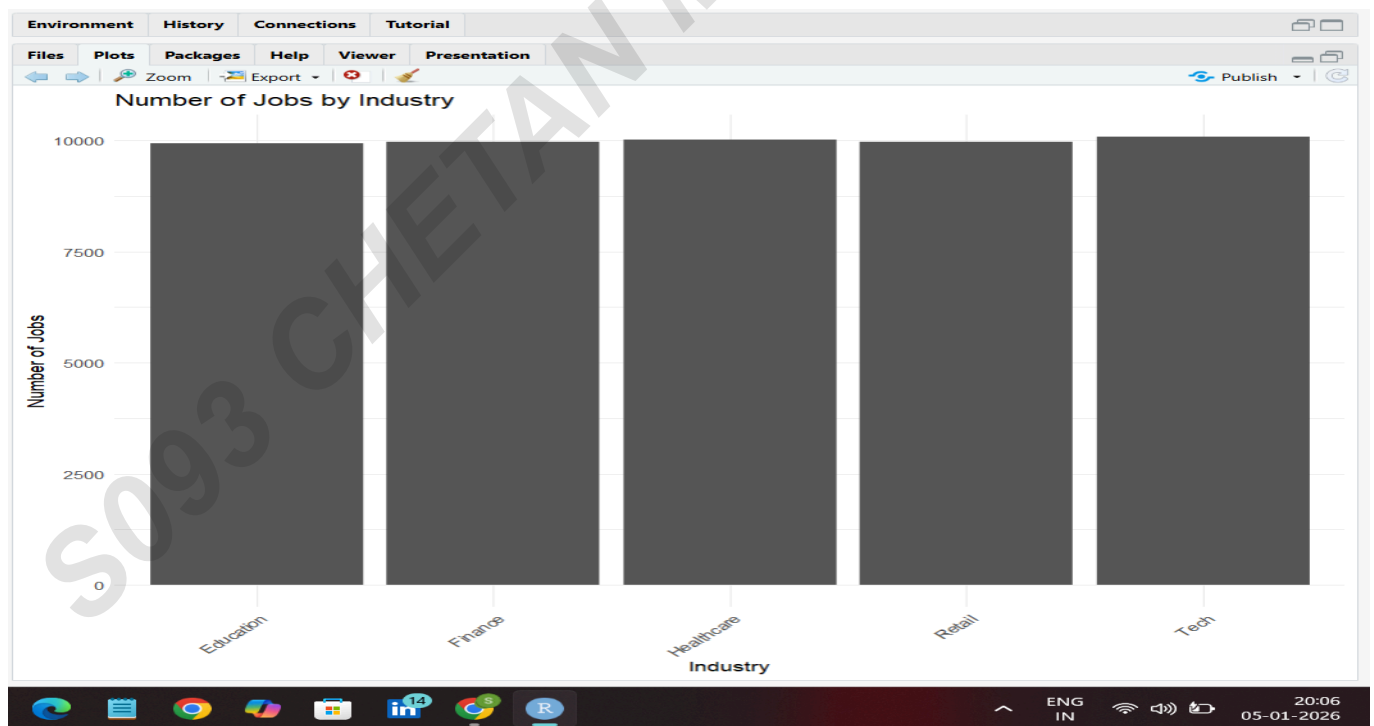
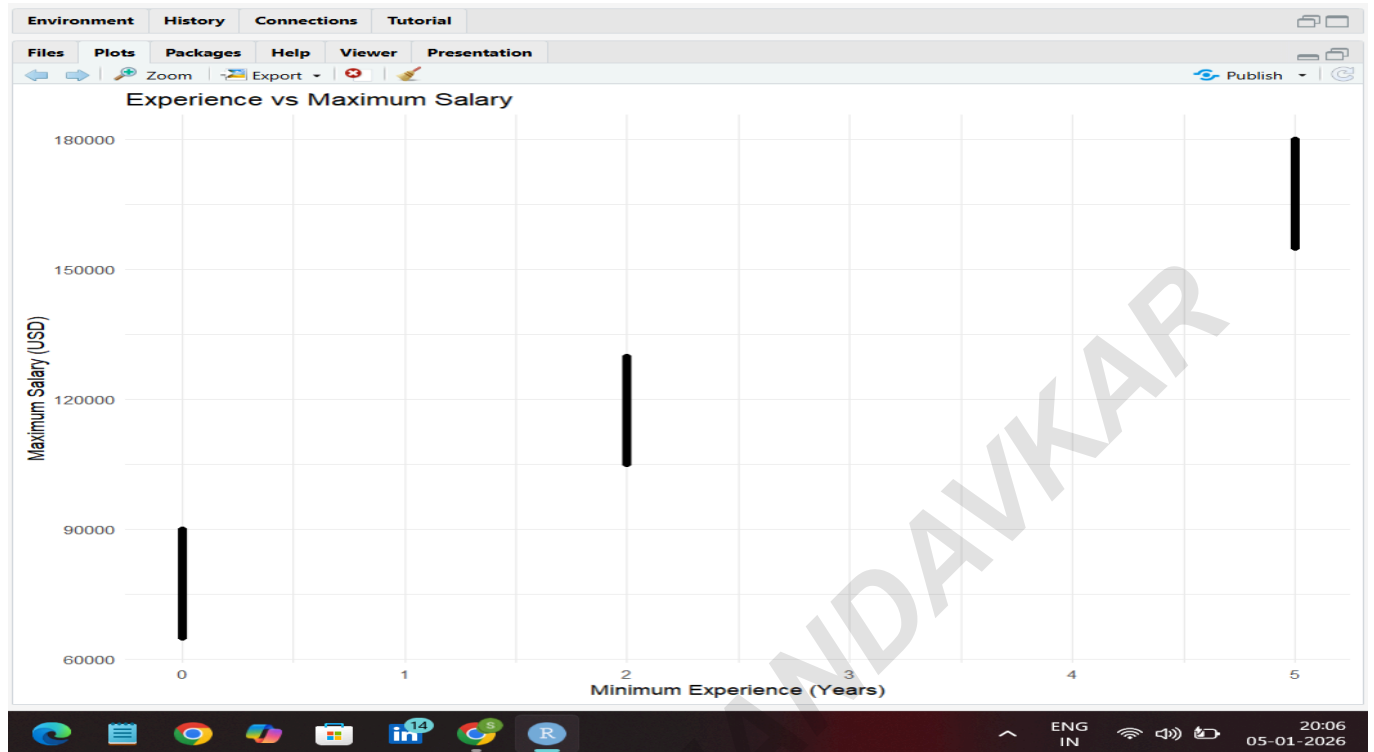


NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE

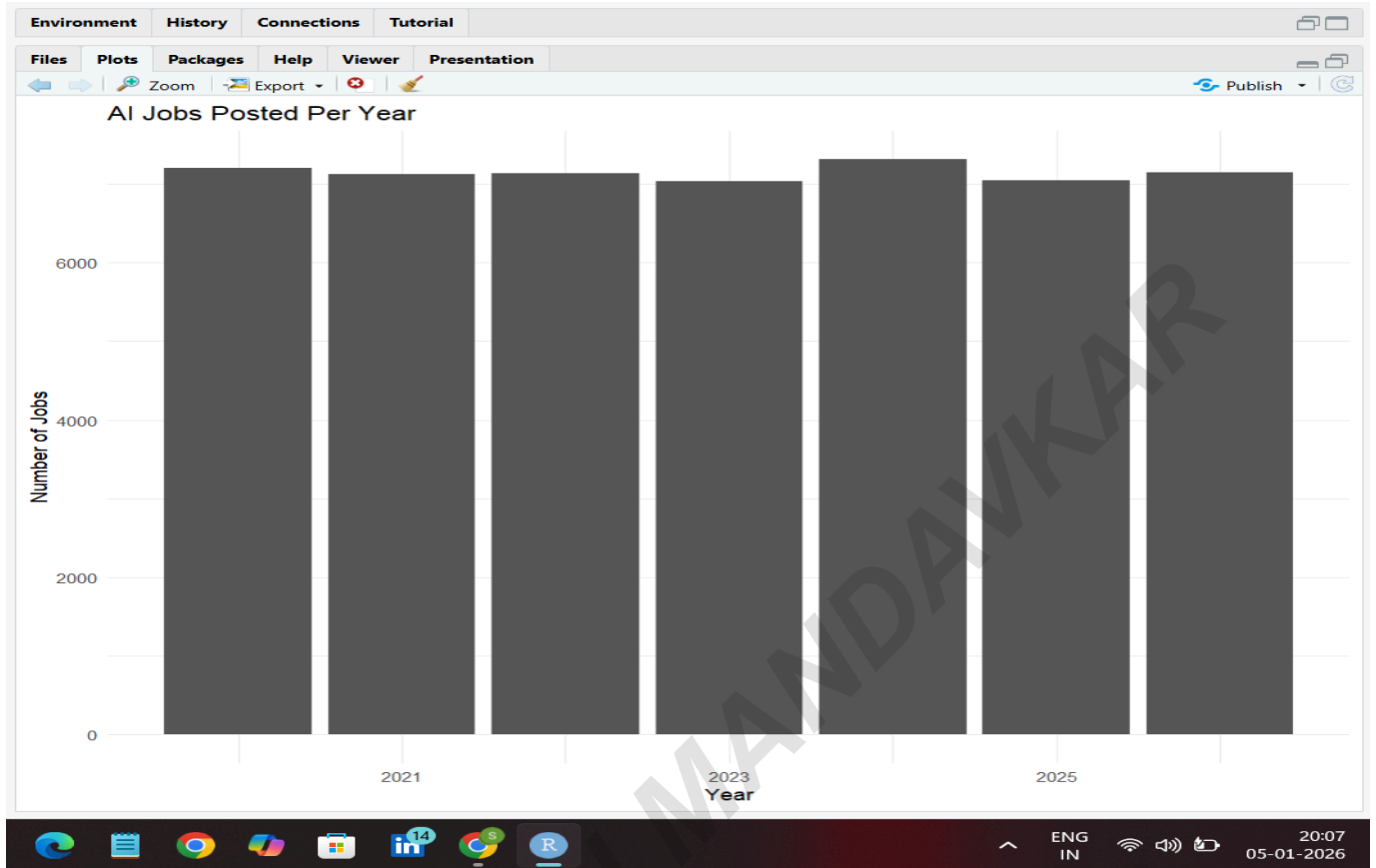


NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE



NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R